

Cosmos — Interactive Architecture

Cloud-Native LLM Orchestration Platform · click any component to learn what it is & why it is here

Flow view

3-D stack

▶ Play the request flow

Tip: click a logo tile for a plain-English detail card · press ▶ to walk the request component-by-component

Client & Edge Presentation API Gateway Control plane Resilience layer Event backbone Worker fleet · Go
LLM providers Storage plane · polyglot Platform & runtime Delivery & supply chain Observability & SRE



LAYER 1 Client & Edge

ingress · TLS · protection



Cloudflare CDN



AWS WAF



Ingress controller



Route 53 · DNS



cert-manager / ACM



LAYER 2 Presentation

dashboard · live view



Next.js · React · TypeScript



SSE live stream



LAYER 3 API Gateway

accept · authenticate · meter



FastAPI gateway



Idempotency + OCC



Keycloak · OAuth2/OIDC/JWT



Quotas + usage ledger



LAYER 4 Control plane

routing · experiments · provisioning



Provider registry & selector



Experimentation / feature flags



Preview-env provisioner (Operator)



LAYER 5 Resilience layer

wraps every provider call



Rate limiter



Circuit breaker



Retry + backoff + jitter



Fallback chain



Load shedding



LAYER 6 Event backbone

durable · decoupled · replayable



Kafka / Redpanda



Schema Registry



Outbox pattern



Dead-letter queue



Debezium · CDC



LAYER 7 Worker fleet · Go

the agent runtime



Go workers



Agent loop



Tool / function calling



Token budget



RAG retrieval



gRPC + Protobuf



Istio service mesh



LAYER 8 LLM providers

the intelligence



Anthropic Claude API



LAYER 9 Storage plane · polyglot

right store for each data shape



PostgreSQL



MongoDB



Redis



pgvector / Qdrant



S3 / MinIO



ClickHouse

▽ SPANS EVERY LAYER ABOVE ▽



CROSS-CUTTING Platform & runtime

where everything runs



Kubernetes · EKS + GKE



Helm + Kustomize



Custom Operator + CRD



KEDA



HashiCorp Vault



CROSS-CUTTING Delivery & supply chain

how code reaches production



GitHub Actions + OIDC



Argo CD · GitOps



Argo Rollouts



Terraform / Crossplane



SBOM · cosign · SLSA



CROSS-CUTTING Observability & SRE

is it healthy? is it fast?



OpenTelemetry



Prometheus



Loki



Grafana



Pyroscope



SLOs + burn-rate alerts



Chaos Mesh



k6

Cosmos · every component annotated with what it is, its role, why chosen, and its trade-offs · logos identify each tool